

Global distribution and dynamics of muddy coasts

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ABSTRACT

Muddy coasts provide important ecological habitats, supply food, and form a natural coastal defence against flooding and erosion. Especially in combination with salt marshes or mangroves they constitute important ecosystem services. In next decades, relative sea level rise, changing coastal wave energy and human interventions will further increase the pressure on muddy coastal zones. For a sustainable, ecosystem-based coastal management it is key to obtain information on the geomorphology and historical changes along muddy areas, that are vulnerable to these pressures. So far, little is known about the distribution and behaviour of muddy coasts at a global scale. In this study we present a global-scale assessment of the occurrence of muddy coasts and rates of coastline change therein. We combine publicly available satellite imagery and coastal geospatial datasets, to train an automated classification method to identify muddy coasts along the world's coastlines. Using a hybrid machine learning algorithm to classify cross-shore coastal transects, we find that 12% of the world's ice-free coastline is muddy, of which 60% is located in the tropics. We found that the majority of the world's muddy coasts are eroding at rates exceeding 0.5 m/yr over the last three decades, while the Ganges delta and Amazone region are retreating at mean rates exceeding -1.3 m/yr.

1 Introduction

Coastal zones, especially along dynamic river deltas, are vulnerable to flooding resulting from storms or coastline erosion resulting from a relative sediment deficit. Many deltaic coastlines have experienced coastal erosion due to trapping of sediment in upstream reservoirs¹² and subsidence³. Globally averaged deltas still experience net land growth over the past 30 years, largely due to larger sediment loads resulting from deforestation⁴. However, in the next decades relative rise in sea level provides an increasingly larger challenge for maintenance of coastlines because many deltas are sinking due to groundwater extraction³ and eustatic sea level rise is rising^{5,6}. On top of relative sea level rise and loss of sediment supply, many coastal areas will also experience increasing mean and extreme coastal wave energy⁷ in the coming decades. Sustainable management of these coastal areas therefore requires thorough understanding of the past and future coastline trends. One way of advancing our understanding of coastline dynamics is by evaluating coastline response on a global scale, providing a methodology to relate coastline dynamics to global changes. Such studies exist for sand-dominated shorelines^{8,9} and for deltas on a more aggregated level⁴. However, despite the global abundance of mud-dominated coastlines, a global inventory of mud coast prevalence and dynamics does not yet exist, probably because the transition from land to water is much more gradual and therefore the location of a muddy shoreline is more difficult to determine.

A muddy open coast is here defined as a coastal depositional environment consisting primarily of fine cohesive sediments formed along an unprotected coastline exposed to low energy conditions¹⁰. These coastal environments are usually associated with rivers, carrying a large sediment load, such as the Amazon River, the Mississippi River and the large Asian rivers draining the Himalayas (notably the Yellow, Yangtze, Red, Mekong, Ayerwaddy, Ganges-Brahmaputra and Indus river deltas). The existence of a muddy open coast requires a supply of mud exceeding its alongshore and cross-shore dispersal rate. On their landward side most mud coasts are flanked by salt marshes (in temperate regions) or mangroves (in the tropics). These ecosystems offer ecological habitats, provide food, and form a natural coastal defence against storms¹¹. Mud coasts, especially in combination with salt marshes or mangroves, therefore constitute important ecosystem services. To better understand the global distribution and the dynamics of mud coasts, we developed a methodology to detect mud-dominated coasts and its dynamics.

Muddy coastlines are more difficult to detect compared to sandy coastlines because the contrast between land and (muddy) water is less pronounced. Therefore we develop a hybrid machine learning method combining classified multispectral satellite imagery with global coastal geophysical datasets. Through supervised machine learning this methodology classifies coastlines into five coastal geomorphological types: 'sandy coasts', 'muddy coasts', 'rocky coasts', 'vegetated coasts' and 'other'. By

sampling large amounts of training and validation data at both pixel (n=3,240) and transect level (n=1,868), we were able to develop an algorithm that identifies muddy coasts with a high accuracy.

Results

Global occurrence of muddy coasts

The hybrid coastal transect classification model accuracy reaches an accuracy of 75% when classifying all five geomorphological types; the addition of geospatial data to the multispectral classification improved the classification accuracy with 11%. Muddy coasts were more accurately classified compared to the average coastal type, with an accuracy of 87%. The hybrid classifier clearly reveals several clusters of muddy coastlines corresponding to well known mud-dominated coastlines (Figure 1). In the America's the most prominent mud coastlines are the Amazon-influenced coastlines¹² and Hudson Bay¹³. The most extensive muddy coastline in Europe is the Wadden Sea¹⁴, whereas the mud-dominated coastlines of west Africa¹⁵ also clearly emerge. However, the most prominent mud-dominated coasts are observed in Asia (particularly the Indian subcontinent, China and Indonesia). This can be explained by the fact that this region receives about 70% of the global suspended sediment load because its sediment yield is much greater than for other drainage basins¹⁶.

Global integration of these findings reveals that 12% of the world's ice-free coastline is muddy, of which 60% is located between 25°N and 25°S. The latter is in accordance with Flemming¹⁷ who estimated that muddy coasts neighbouring mangrove systems in this region cover an area of about 75%. The latitudinal distribution is consistent with the distribution of Hayes¹⁸ and reflecting the large suspended load carried by tropical rivers¹⁶. The longitudinal distribution reflects the large suspended loads carried by rivers draining the Himalayan and southeast Asian rivers, as well as the Amazon river.

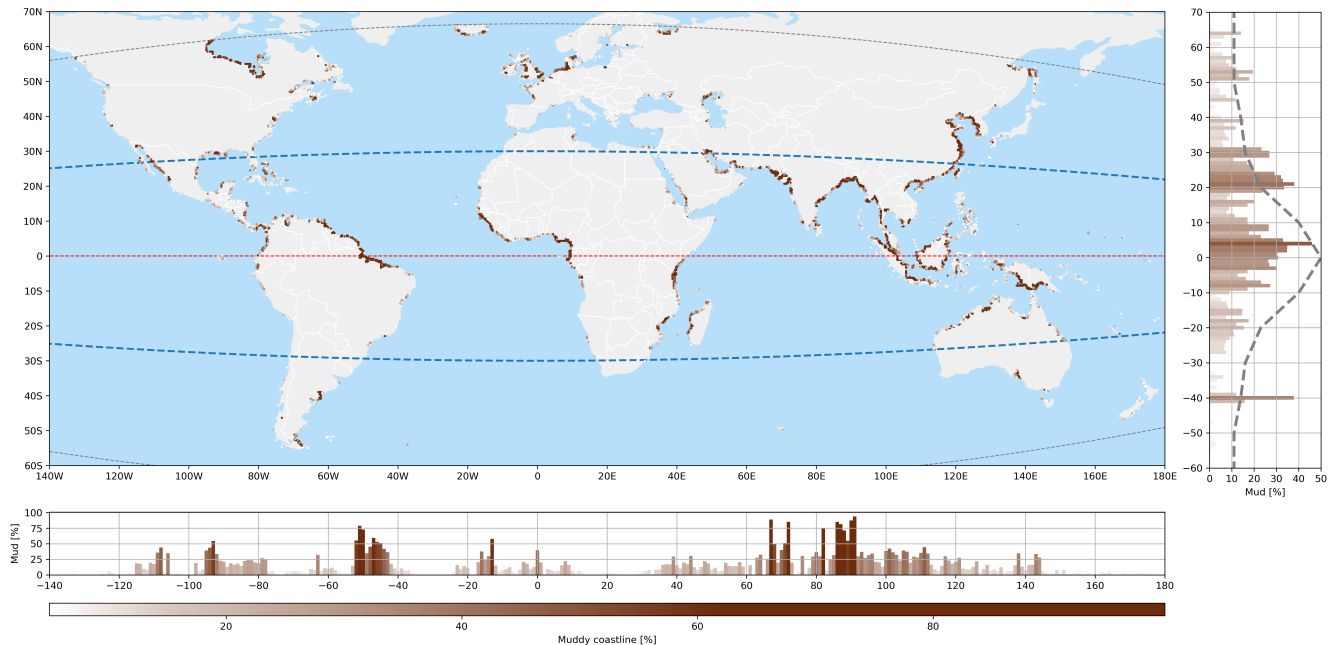


Figure 1. Global distribution of muddy coastlines derived from the hybrid classification model, with colours denoting the local percentage of mud (darker brown implying more mud). The subplot on the right averages the relative occurrence of muddy coastlines per latitudinal degree, with the dashed line representing the distribution of muddy coastlines reported by Hayes¹⁸. The lower subplot shows the relative occurrence of muddy coastlines per longitudinal degree. The curved grey dashed lines in the main plot indicate the boundaries of the ice-free coastlines considered in this analysis.

Dynamics of muddy coasts vs. sandy coasts

Globally, sandy coastline are relatively stable, with approximately as much eroding beaches as accreting beaches⁸. Although the global dynamics of undifferentiated coastlines (without distinguishing between the coastline type) have been investigated (Mentaschi2018GlobalAccretion and nienhuis2020globalb) a methodology to differentiate between the dynamics of muddy coasts and sandy coasts does not yet exist. This is important, because the processes driving their long-term stability, and the role of human interventions as well as climate change, is very different for muddy coastlines than for sandy coastlines. Many river deltas are mud-dominated and relatively flat, and therefore constitute vast low-lying areas which are sensitive to the delicate

balance between sea level rise, land subsidence and sedimentation¹⁹. To derive the historical changes of muddy coastlines, we adopted the method of Luijendijk *et al.*⁸ for the period 1984–2016. This methodology was developed and validated for sandy beaches, requiring slight adaptations for application to mud-dominated coastlines (see SM). After filtering the muddy coastline transects for a minimum number of historic data points, temporal coverage, potential ice coverage and outliers, 97.7% of the mud coasts remain suitable for further analysis.

The analysis of coastline dynamics reveals that 40% of the world's muddy coasts are stable (absolute long-term change rates of less than 1 m/yr over the period 1984–2016); 31% are persistently eroding, while 29% are accreting (Table 1). About 7.8% of the muddy coasts exceed erosion rates of more than 10 m/yr, while 8.8% is accreting at rates more than 10 m/yr. On a global scale, the world's muddy coasts have accreted on average 0.18 m/yr over the three decades of the study period, i.e. a total gain of 579 km² over this period. *** Arjen, can you provide a similar number for sandy beaches? *** Muddy coastlines are generally more dynamic than sandy coasts, reflected in much more (very) extreme accretion and erosion. Rapid accretion rates can be explained by the dominance of fine-grained suspended load in the world's major river systems¹⁶; rapid erosion rates by the low gradient of mud-dominated coastal environment (resulting in large horizontal erosion rates for a given vertical erosion rate).

Dynamic class	Rate	Sandy coasts	Muddy coasts
Very extreme accretion	> 10 m/yr	1.6%	8.8%
Extreme accretion	5 to 10 m/yr	1.8%	5.4%
Severe accretion	3 to 5 m/yr	2.3%	4.3%
Intense accretion	1 to 3 m/yr	10.1%	10.7%
Stable	-1 to 1 m/yr	69.7%	40.3%
Intense erosion	-1 to -3 m/yr	9.8%	12.2%
Severe erosion	-3 to -5 m/yr	1.9%	4.8%
Extreme erosion	-5 to -10 m/yr	1.5%	5.7%
Very extreme erosion	< -10 m/yr	1.2%	7.8%

Table 1. Chronic coastline dynamics classification scheme.

Mapping of muddy areas experiencing rapid accretion and erosion reveal several clusters of erosion and of sedimentation (Figure 2). Eroding coastlines may be the result of a reduction in the sediment load^{1,2,4}, subsidence³, expansion of a high natural sediment load or a more recent increase resulting from deforestation⁴. Areas at higher latitude as well as along China's coastline are typically expanding. The expansion of China's coastline is the result of the large sediment loads carried by the Yellow and Yangtze rivers (carrying the second and fourth largest sediment load²⁰). The expansion of muddy coasts in North America's Hudson Bay is likely the result of isostatic rebound leading to sealevel fall²¹; the expansion in Europe is related to land reclamations²². Many coastlines at lower latitudes experience erosion (North America, Pakistan, India) or alternate between erosion and sedimentation (the Amazon coastline, Indonesia, Bangladesh). Erosion along the Pakistan coastline is primarily driven by a reduction of the sediment load of the Indus (95%²³). The alternating patterns of erosion and accretion along the northeast coast of South America reflects the natural migration cycles of mud banks^{24,12}.

Two out of the five areas with strongest erosion rates are located in Bangladesh (Table 2), where stretches of 14 to 31 km long muddy coasts have eroded on average between 20 and 50 m/yr over the study period. The world's longest muddy coastal stretch suffering severe erosion is located in the south west in Papua New Guinea where we observed a 45 km stretch of muddy coast (the banks of a river mouth) with a mean erosion rate of 14.0 m/yr. The largest accretive hot spot is in Indonesia, North Sumatra at the straits of Malacca, at the river mouth of the Rokan River, where a stretch of 17.5 km of muddy coast has accreted on average 37.6 m/yr. Not only the mudflats seem to have accreted, but also the (mangrove) forest / vegetation has expanded seaward. It is noteworthy that at the north east coast of South America large erosion and accreting hot spots alternate. The Guyana current transports Amazonian sediments (mainly mud sediments) along the north coast of Brazil into French Guyana and Suriname²⁵, where mudbanks migrate westward and constantly change shape and orientation due to the combined action of waves and currents²⁶.

From a continental perspective, Africa, Oceania and South America are continents subject to net erosion (-0.14 m/yr, -0.44 m/yr and -0.76 m/yr, respectively) whereas the other three continents accrete (table in Figure 2). The continents in the northern hemisphere all accrete on average while the continents in the southern hemisphere are, on average, eroding.

Methods

Our hybrid coastal transect classification model (Figure 3) maps the output of a multispectral pixel-based classifier and global geospatial data on a global coastal transect system⁸. These cross-shores transects are 1500 m long and placed perpendicular to

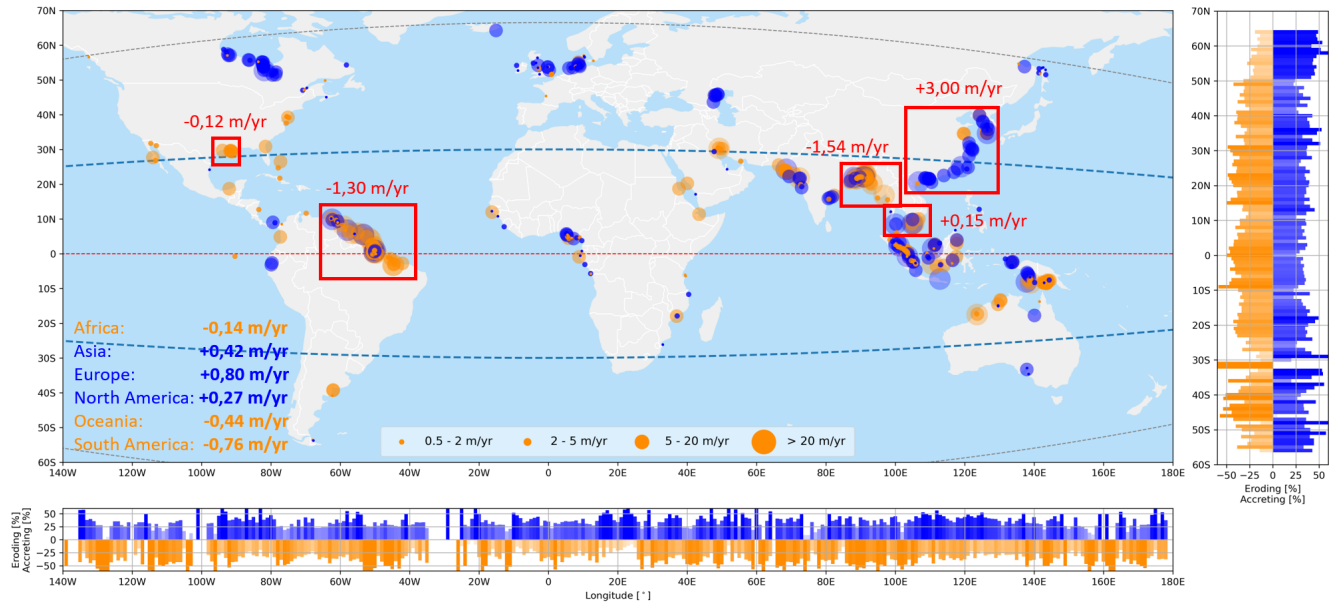


Figure 2. Global hot spots of muddy coast erosion and accretion; the orange (blue) circles indicate erosion (accretion) for the four relevant coastline dynamic classifications (see legend). The bar plots to the right and at the bottom present the relative occurrence of eroding (accreting) muddy coastlines per degree latitude and longitude, respectively.

the 2016 global OpenStreetMap (OSM) coastline¹. The spacing of each transect decreases from approximately 500 m close to the equator to 200 m closer to the poles (66.5°N and 66.5°S), resulting in total in 1.8 million transects. In the following sections, we will first explain the pixel-based classification and combination with global datasets into the hybrid classifier, followed by a validation of this methodology.

Pixel-based multispectral classifier

The pixel-based multispectral image classifier uses freely available Sentinel-2 (S2) multispectral global Top of Atmosphere (TOA) images², to classify coastal-cover and -use of the global coastal area. The S2 Multispectral Instrument (MSI) samples thirteen spectral bands: visible and near infrared (NIR) at a spatial resolution of 10 m, red edge (RE) and short wave infrared (SWIR) at 20 m, and atmospheric bands at 60 m. The water vapour band (B9) and cirrus band (B10) were excluded from the analysis as these bands are most sensitive to moisture in the atmosphere.

First, we collect all TOA reflectance imagery of the year 2020 which contained less than 15% cloud cover to compile a one year composite image. This composite image was analysed to extract spectral reflectance properties of different coastal classes and to remove tidal and seasonal variability in wave, beach and vegetation characteristics (following Hansen *et al.*²⁷ and Luijendijk *et al.*⁸). As a next step, subsections of the global pixel distribution are calibrated. For this purpose the global coastal area was divided into approximately 24,000 square boxes of 20 x 20 km. 75 of these 24,000 boxes, representing different coastal environments on the six continents (excluding Antarctica), were selected for training. In total 3,240 training data points were manually selected and labelled with a coastal class through visual inspection of the RGB 2020 S2 composite TOA images. The training data consisted of seven clearly distinguishable coastal classes: (1) sandy beaches, (2) mudflats, (3) clear water, (4) turbid (brown) water, (5) green vegetation, (6) dry vegetation, and (7) other (containing clouds, snow, buildings and all other indefinable points). An optimized²⁸ random forest²⁹ ensemble classifier, directly available in Google Earth Engine³⁰ as SmileRandomForest, was trained to the spectral reflectance properties of 3,240 labelled points. The algorithm was implemented using 15 decision-trees, while other parameters, such as the bagging fraction, were left as default. When trained, the algorithm is able to assign a class label (i.e. sand, mud, water, turbid water, vegetation, dry vegetation, other) to each of the pixels in the multispectral satellite imagery. The global multispectral image classifier is verified and validated using sediment samples in a Dutch estuary, which is discussed in Section Performance.

¹<https://planet.osm.org/>

²<https://scihub.copernicus.eu/>

	Areal change rate (m ² /yr)	Mean change rate (m/yr)	Length of section (km)
Erosive hot spot muddy coasts			
Charfasson, Bangladesh	-960,313	-31.0	31
Pulau Beruit, Malaysia	-773,040	-19.3	40
Johanna Maria, Suriname	-771,321	-55.1	14
Ilha de Maracá, Brazil	-748,139	-37.4	20
Sandwip Island, Bangladesh	-709,151	-50.7	14
Accretive hot spot muddy coasts			
Sinaboi, Indonesia	657,898	37.6	17.5
Đồng Hưng, Tiên Lãng, Vietnam	579,164	41.4	14
Viên An, Ngoc Hien District, Vietnam	494,692	41.2	12
Huian, Quanzhou, China	449,099	26.4	17
North Coast of Friesland, Netherlands	441,492	13.0	34

Table 2. World's largest erosive and accretive muddy coast hot spots.

Global coastal geospatial data

In parallel to the multispectral classifier, we also analysed the physical geospatial characteristics using global coastal datasets. We selected seven freely available coastal geophysical datasets that represent different characteristics (Table 3).

Coastal data type	Coastal geospatial dataset	Spatial resolution	Reference
Elevation	MERIT Digital Elevation Model	90 m	Yamazaki <i>et al.</i> ³¹
Temperature	WorldClim V1 Bioclim Dataset	1,000 m	Hijmans <i>et al.</i> ³²
Mangroves	Global Mangrove Forests Distribution, 2000	30 m	Giri <i>et al.</i> ³³
Tidal flat width	Global Intertidal Change Dataset	30 m	Murray <i>et al.</i> ¹¹
Transition zone width	Global Surface Water dataset	30 m	Pekel <i>et al.</i> ³⁴
Tidal range	Deltares Global Tide and Surge Model	5,000 m	Verlaan <i>et al.</i> ³⁵

Table 3. Overview of global coastal geospatial datasets used in this study

The Multi-Error-Removed Improved-Terrain Digital Elevation Model (MERIT DEM)³¹ provides indirect information on the coastal environment. A high and variable bed level represents a rocky coast, while a low (variable) coastal topographic relief is usually related to a river delta and therefore to a supply of sediment to the coasts. The majority of mud particles are formed by chemical weathering, and therefore muddy coasts are most abundant in the humid tropical regions. Climatological conditions were extracted from the WorldClim V1 Bioclim dataset³². The presence of mangrove forests is also a strong indicator for the presence of mud, and is obtained from the Global Mangrove Forests Distribution³³. Tidal flats are common in sediment-rich environments where the tidal range is large relative to the typical wave height³⁶. Tidal flats may range from sand-dominated to mud-dominated, but are typically a combination with mud-dominated upper flats coarsening in the seaward direction. The occurrence of tidal flats can be estimated from global tidal flat ecosystem maps¹¹ and the Global Surface Water (GSW) dataset³⁴, which can be interpreted as the intertidal area where dry and wet areas alternate (transition zone). Since the tidal range is also a strong indicator for tidal flats occurrence, we also include a global tidal dataset (the Deltares Global Tide and Surge Model³⁵ (GTSM)).

The coastal geospatial data for elevation, mangroves, tidal flat width and transition zone width is extracted every 10 m along the transects (same spatial resolution as the S2 pixels). A number of scalars are subsequently computed per transect. The elevation data per transect is converted into a maximum bed level and a variance along the profile. The Global Mangrove Forests Distribution dataset is used to determine whether a transect crosses mangrove forests (True/False), independent of the amount of mangroves. The intertidal area dataset is used to compute the tidal flat width along each transect. A water probability profile is drawn along each transect using the GSW dataset, from which the width of the transition zone of land to water is calculated; this transition zone is bounded by 5% to 95% water probability in order to remove the (nearly) converging start and end of the profile. The resolution of the temperature and tidal range data is coarser than the resolution of transects, and

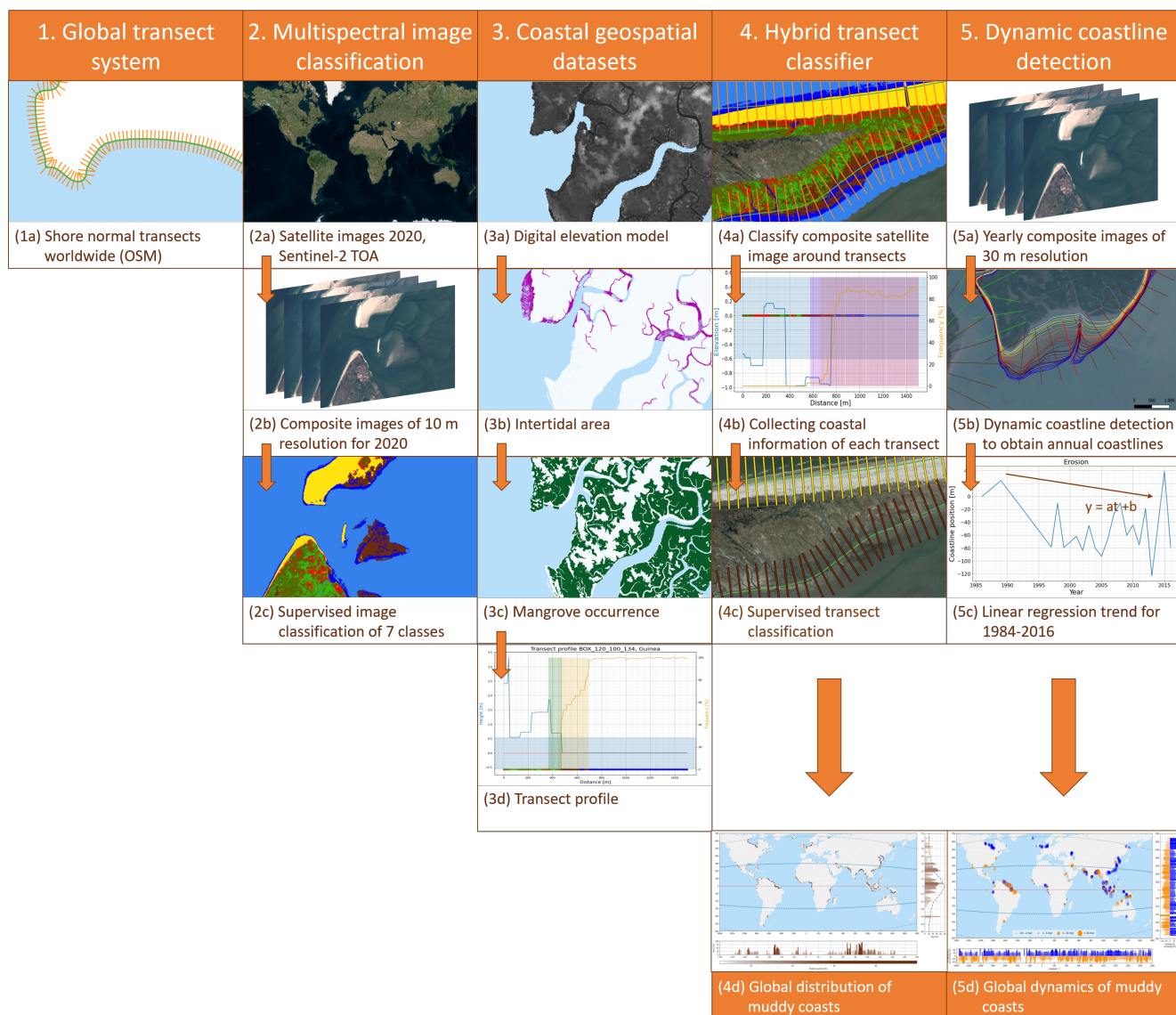


Figure 3. The procedure followed for classifying the global coastline using a global transect system, pixel-based multispectral classification and coastal geospatial data.

therefore the temperature and the tidal range data is interpolated using a nearest neighbour interpolation technique. Per transect the maximum temperature of the warmest month and the minimum temperature of the coldest month are extracted from the temperature dataset. The tidal range is calculated per transect by subtracting the mean lower low water from the mean higher high water. Finally, the longitude and latitude of each transect is defined by the coordinates at the centroid of that transect.

The hybrid transect classifier: detection of muddy coasts

Both the spectral reflectance properties of mud at pixel level provided by the multispectral satellite images and the physical geospatial characteristics collected with the global coastal geospatial datasets are subsequently integrated into a hybrid transect classifier. The principle of the hybrid transect classifier is illustrated with two examples located in Madagascar and in Suriname, see Figure 4. The left-hand panels of the figure depict the 2020 S2 composite TOA image, with five transects each classified by the multispectral image classifier. From the classified transects we obtained seven scalars representing the presence of the seven coastal classes along each transect. The right-hand panels of the figure provide indicators from the coastal geospatial datasets in combination with the results of the multispectral classifier. This visualisation allows for an intuitive, contextual inspection of the available information per coastal transect.

To finally train the hybrid coastal transect classification model that can recognise muddy coasts at global scale, we created

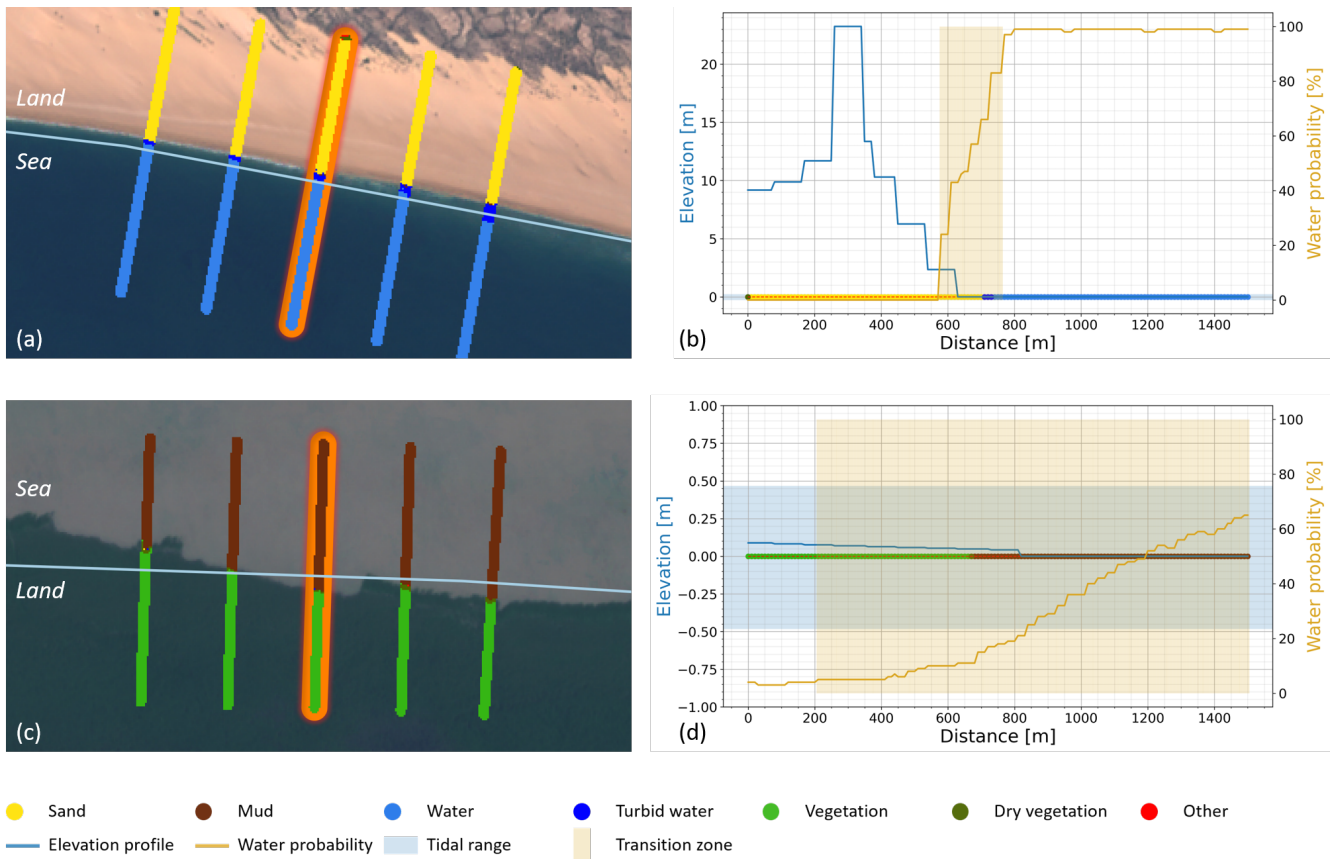


Figure 4. Upper panels: Madagascar; bottom panels: Suriname. Left-hand panels: 2020 S2 composite TOA images with five transects (1500 m in length) each classified by the multispectral image classifier. Right-hand panels: transect profiles of the orange marked transect from the left-hand panel. The horizontal axis is the distance [m] along the transect, where 0 m is the landward end and 1500 m the seaward end. The left y-axis represents the elevation [m] above mean sea level to which the elevation profile and the tidal range are related, the right y-axis represents the water probability [%]. The multispectral image classification is shown at 0 m elevation in the graphs with coloured dots referring to the coastal class of each pixel along the transect. The tidal range is plotted as a horizontal shaded bar, the transition zone is plotted as a vertical shaded bar.

a transect training dataset consisting of 1,868 manually labelled transects. These training transects are randomly distributed around the globe and capture different coastal environments under different conditions. The training transects were labelled by visual inspection of S2 TOA composite RGB images of 2020. Five coastal types have been defined: (1) sandy coasts (a clear strip of sandy sediment at the coastline is visible on the satellite image); (2) muddy coasts (an exposed tidal flat/mud bank is visible on the satellite image); (3) rocky coasts (large rocky features/cliffs are visible on the satellite image); (4) vegetated coasts (vegetation covers the coastline and no sediment/beach is visible between water and land); and (5) other (all indefinable coasts, including coasts covered with ice/snow, anthropogenic structures that shape the coastline such as harbours and dykes). The transect training dataset consisted of a (nearly) equal amount of transects per coastal type.

The processed information of the labelled training transects provides a vector, which entries are individual coastal features (i.e. the presence of sand pixels along the transect or the maximum elevation of the topographic profile, etc.), for training the one dimensional hybrid coastal transect classification model, again, using a SmileRandomForest algorithm, with similar configurations as the multispectral image classifier. The output of the classifier assigned each transect a predicted label with the corresponding coastal type (i.e. sandy coast, muddy coast, rocky coast, vegetated coast, other) that best matched the observed coastal characteristics of the transect. With this classifier all global coastal transects were classified into a certain coastal type.

Performance

Training and accuracy

As explained above, 25% of the 3,240 manually labelled training points are used for validation, and 75% for training. The overall accuracy of the image classification model increases with the number of training points, reaching 80% with 500 training

points and is maximally 83% with all training points, with the 90% confidence intervals at $\pm 2\%$. The f1-score strongly varies with the coastal class, with water and sand being more accurately resolved than vegetation, mud, or turbid water.

Similar to the training and validation of the pixel land-type classifier, 75% of the 1,868 manually labelled training transects are used for training, and 25% for validation. Using the validation transects, we evaluate the accuracy of the multispectral classification method, the geospatial classification method and the hybrid classification method. The highest overall accuracy is achieved with the hybrid transect classifier (see left-hand panel in Figure 5): 75% accuracy with a 90% confidence interval of $\pm 2\%$. The addition of geospatial data to the multispectral classification significantly improved (+11%) the hybrid coastal transect classification model accuracy. The accuracy of all three transect classifiers are still increasing with the number of training transects, implying that the accuracy of classification may be improved by adding more labelled transect training data. The f1-scores reveal that transects are most accurately classified for muddy coasts, and that adding the geospatial data especially improves the classification of rocky and vegetated coasts (see SM).

The hybrid transect classifier uses a total of seventeen features of which the three most important originate from the multispectral classifier (water, sand and mud) whereas the following three in order of importance (maximum elevation, elevation variance and transition zone width) are based on the geospatial datasets (see right-hand panel in Figure 5). The way the classifier is influenced by similar coastal types is visualised in a confusion matrix (see SM). The coastal type that interferes most with muddy coasts are vegetated coasts: although 92 transects (78%) classified as mud coasts were correctly classified, 13 transects (11%) were in reality vegetated (and 7 sandy (6%)). This is probably because the geospatial characteristics of these coasts are very similar, with the exception of (sometimes narrow) mud banks visible on multispectral images. However, 95% of the actual muddy transects were correctly identified by the classifier.

The typical physical geospatial features are pronounced for the different coastal environments (see right-hand panels in Figure 4; top: sandy coast, bottom: muddy coast). Sandy coastlines, for instance, have a larger maximum elevation and narrower transition zone compared to a muddy coastline. These features can therefore be used to optimise the classification. To this end the density distribution of the muddy, sandy, rocky and vegetated coasts are visualised against the geospatial features using the 1,868 labelled training transects (see SM). Muddy coasts generally present a low maximum elevation profile, with a narrow density distribution and a peak at 5 m. The spread of maximum elevation becomes progressively larger for sandy coasts (with a peak at 10 m), vegetated coasts and rocky coasts. Muddy coasts have the widest transition zones, often occupying half the length of the transect. The transition zone width is much narrower for the sandy coasts, rocky coasts and vegetated coasts.

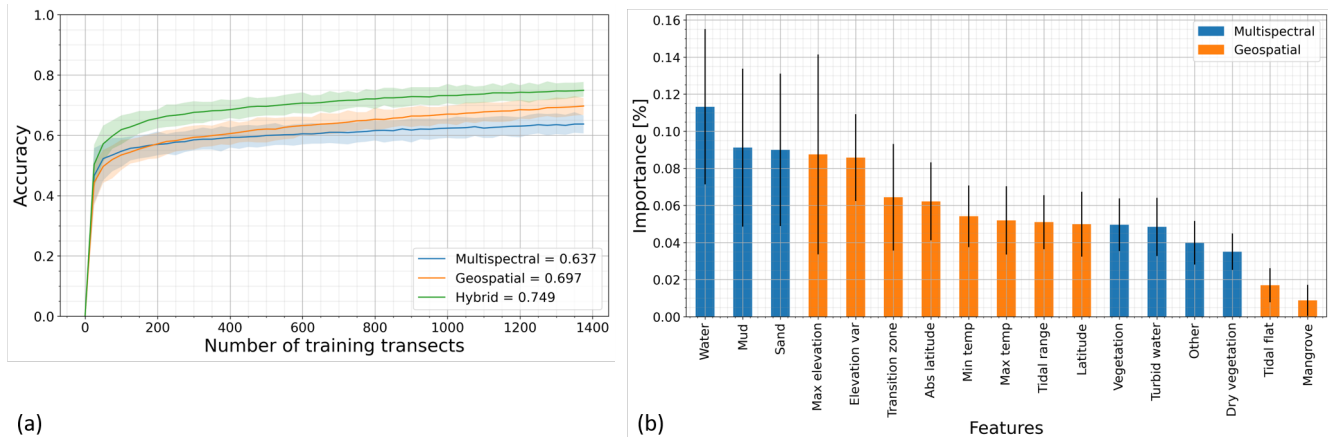


Figure 5. Left-hand panel: overall accuracy for three transect classifiers (multispectral, geospatial and hybrid) that classify coastal transects into five coastal types. The number of used training transects is on the x-axis and the accuracy on the y-axis. The solid line represents the mean accuracy, the shaded area represents the 90th percentile of the accuracy. Right-hand panel: importance of features used in the hybrid transect classifier in descending order. The blue (orange) bars represent the mean of the importance for each feature originating from the multispectral satellite images (coastal geospatial datasets). The black lines represent the standard deviation of the importance of each feature.

Ground validation

The pixel-based image classification is subsequently validated against in-situ sediment observations of sediment grain size distribution in the Dutch Wadden Sea (the Sediment Atlas³). Although these data were collected in the period 1989 - 1995, the distribution of sand and silt has changed only to a very limited extent³⁷. We assume that soil samples with more than 25%

³https://puc.overheid.nl/rijkswaterstaat/doc/PUC_43681_31/

mud can be classified as muddy (corresponding to the approximate transition from a non-cohesive sediment bed to a cohesive sediment bed³⁸). The exposed coastlines of the barrier islands of the Wadden Sea are sandy, whereas the interior is muddy, which is in high agreement with observations - see also Alonso *et al.*³⁷). Our analysis reveals that pixels classified as sand generally have a low mud content: 99% of the pixels classified as sandy have a mud content below 25%. However, the amount of pixels classified as mud is too large: only 64% of the pixels classified as muddy have a mud content exceeding 25%. This overestimation is the result of a strip of turbid water sometimes separating the beach and clear water, which is incorrectly classified as a muddy coastline. So even though these coasts should be formally classified as sandy (instead of muddy) they are still mixed sand-mud shores (rather than a purely sandy coast).

Dynamic coastline detection

After the classification of all global coastal transects, we correlated the classified muddy transects with long-term changes in the Luijendijk *et al.*⁸ dataset. To avoid unrealistic coastline change rates we applied similar filters as in Luijendijk *et al.*⁸ to all muddy transects. The muddy transect data is filtered on the number of Satellite Derived Shoreline (SDS) data points and the temporal coverage per transect. Transects with possible ice coverage are discarded. Outliers in the SDS are excluded from the linear regression.

The performance of the linear regression method, used to quantify long-term coastline change rates, in capturing trends of chronic muddy coastline change, is assessed (see SM). The uncertainty bandwidth, which can be considered as a proxy for the representativeness of the linear regression method to express the long-term change rates, is analysed for the muddy transects. This analysis showed that the uncertainties are of the same order of magnitude as for the coastline dynamics of sandy coasts presented in Luijendijk *et al.*⁸. Even more, the stronger change rates of the muddy transects are more accurate in terms of confidence intervals. We therefore conclude that the adaption of the long-term coastline changes for muddy coasts is justified.

References

1. Syvitski, J. P. M., Vorosmarty, C. J., Kettner, A. J. & Green, P. Impact of humans on the flux of terrestrial sediment to the global coastal ocean. *science* **308**, 376–380 (2005).
2. Vörösmarty, C. J. *et al.* Anthropogenic sediment retention: major global impact from registered river impoundments. *Glob. planetary change* **39**, 169–190 (2003).
3. Syvitski, J. P. M. *et al.* Sinking deltas due to human activities. *Nat. Geosci.* **2**, 681–686 (2009).
4. Nienhuis, J. H. *et al.* Global-scale human impact on delta morphology has led to net land area gain. *Nature* **577**, 514–518 (2020).
5. Nicholls, R. J. & Cazenave, A. Sea-level rise and its impact on coastal zones. *science* **328**, 1517–1520 (2010).
6. Bilskie, M. V., Hagen, S. C., Medeiros, S. C. & Passeri, D. L. Dynamics of sea level rise and coastal flooding on a changing landscape. *Geophys. Res. Lett.* **41**, 927–934, DOI: [10.1002/2013GL058759](https://doi.org/10.1002/2013GL058759) (2014).
7. Li, N., Yamazaki, Y., Roeber, V., Cheung, K. F. & Chock, G. Probabilistic mapping of storm-induced coastal inundation for climate change adaptation. *Coast. Eng.* **133**, 126–141, DOI: [10.1016/j.coastaleng.2017.12.013](https://doi.org/10.1016/j.coastaleng.2017.12.013) (2018).
8. Luijendijk, A. *et al.* The State of the World's Beaches. *Sci. Reports* **8**, 1–11, DOI: [10.1038/s41598-018-24630-6](https://doi.org/10.1038/s41598-018-24630-6) (2018).
9. Mentaschi, L., Voudoukas, M. I., Pekel, J. F., Voukouvalas, E. & Feyen, L. Global long-term observations of coastal erosion and accretion. *Sci. Reports* **8**, 1–11, DOI: [10.1038/s41598-018-30904-w](https://doi.org/10.1038/s41598-018-30904-w) (2018).
10. Healy, T. R., Wang, Y. & Healy, J.-A. *Muddy coasts of the world : processes, deposits, and function* (Elsevier, 2002).
11. Murray, N. J. *et al.* The global distribution and trajectory of tidal flats. *Nature* **565**, 222–225, DOI: [10.1038/s41586-018-0805-8](https://doi.org/10.1038/s41586-018-0805-8) (2019).
12. Anthony, E. J. *et al.* The Amazon-influenced muddy coast of South America: A review of mud-bank–shoreline interactions. *Earth-Science Rev.* **103**, 99–121 (2010).
13. Stewart, D. B. & Lockhart, W. L. *An overview of the Hudson Bay marine ecosystem* (Department of Fisheries and Oceans, Central and Arctic Region, 2005).
14. Straaten, L. M. J. U. *Composition and structure of recent marine sediments in the Netherlands* (1954).
15. Anthony, E. J. The muddy tropical coast of West Africa from Sierra Leone to Guinea-Bissau: geological heritage, geomorphology and sediment dynamics. *Afr. Geosci. Rev* **13**, 227–237 (2006).
16. Milliman, J. D. & Meade, R. H. World-wide delivery of river sediment to the oceans. *The J. Geol.* **91**, 1–21 (1983).
17. Flemming, B. W. Geographic distribution of muddy coasts (2002).

18. Hayes, M. O. Relationship between coastal climate and bottom sediment type on the inner continental shelf. *Mar. Geol.* **5**, 111–132, DOI: [https://doi.org/10.1016/0025-3227\(67\)90074-6](https://doi.org/10.1016/0025-3227(67)90074-6) (1967).
19. Milliman, J. D., Broadus, J. M. & Gable, F. Environmental and economic implications of rising sea level and subsiding deltas: the Nile and Bengal examples. *Ambio* **18**, 340–345 (1989).
20. Milliman, J. D. & Syvitski, J. P. M. Geomorphic/tectonic control of sediment discharge to the ocean: the importance of small mountainous rivers. *The journal Geol.* **100**, 525–544 (1992).
21. Sella, G. F. *et al.* Observation of glacial isostatic adjustment in “stable” north america with gps. *Geophys. Res. Lett.* **34** (2007).
22. Van Maren, D., Oost, A., Wang, Z. B. & Vos, P. The effect of land reclamations and sediment extraction on the suspended sediment concentration in the ems estuary. *Mar. Geol.* **376**, 147–157 (2016).
23. Syvitski, J. P. *et al.* Anthropocene metamorphosis of the indus delta and lower floodplain. *Anthropocene* **3**, 24–35 (2013).
24. Allison, M. A. & Lee, M. T. Sediment exchange between Amazon mudbanks and shore-fringing mangroves in French Guiana. *Mar. Geol.* **208**, 169–190, DOI: [10.1016/j.margeo.2004.04.026](https://doi.org/10.1016/j.margeo.2004.04.026) (2004).
25. Augustinus, P. G. Actual development of the chenier coast of suriname (South America). *Sedimentary Geol.* **26**, 91–113, DOI: [10.1016/0037-0738\(80\)90007-X](https://doi.org/10.1016/0037-0738(80)90007-X) (1980).
26. de Vries, J., van Maanen, B., Ruessink, G., Verweij, P. A. & de Jong, S. M. Unmixing water and mud: Characterizing diffuse boundaries of subtidal mud banks from individual satellite observations. *Int. J. Appl. Earth Obs. Geoinformation* **95**, 102252, DOI: [10.1016/j.jag.2020.102252](https://doi.org/10.1016/j.jag.2020.102252) (2021).
27. Hansen, M. C. *et al.* High-Resolution Global Maps of 21st-Century Forest Cover Change. *Science* **342**, 850–853, DOI: [10.1126/science.1244693](https://doi.org/10.1126/science.1244693) (2013).
28. Li, H. Smile. <https://haifengl.github.io> (2014).
29. Breiman, L. Random Forests. *Mach. Learn.* **45**, 5–32, DOI: [10.1023/a:1010933404324](https://doi.org/10.1023/a:1010933404324) (2001).
30. Gorelick, N. *et al.* Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote. Sens. Environ.* **202**, 18–27, DOI: [10.1016/j.rse.2017.06.031](https://doi.org/10.1016/j.rse.2017.06.031) (2017).
31. Yamazaki, D. *et al.* A high-accuracy map of global terrain elevations. *Geophys. Res. Lett.* **44**, 5844–5853, DOI: [10.1002/2017GL072874](https://doi.org/10.1002/2017GL072874) (2017).
32. Hijmans, R. J., Cameron, S. E., Parra, J. L., Jones, P. G. & Jarvis, A. Very high resolution interpolated climate surfaces for global land areas. *Int. J. Climatol.* **25**, 1965–1978, DOI: [10.1002/joc.1276](https://doi.org/10.1002/joc.1276) (2005).
33. Giri, C. *et al.* Status and distribution of mangrove forests of the world using earth observation satellite data. *Glob. Ecol. Biogeogr.* **20**, 154–159, DOI: [10.1111/j.1466-8238.2010.00584.x](https://doi.org/10.1111/j.1466-8238.2010.00584.x) (2011).
34. Pekel, J. F., Cottam, A., Gorelick, N. & Belward, A. S. High-resolution mapping of global surface water and its long-term changes. *Nature* **540**, 418–422, DOI: [10.1038/nature20584](https://doi.org/10.1038/nature20584) (2016).
35. Verlaan, M., De Kleermaeker, S. & Buckman, L. GLOSSIS: Global Storm Surge Forecasting and Information System. In *In: Australasian Coasts & Ports Conference 2015: 22nd Australasian Coastal and Ocean Engineering Conference and the 15th Australasian Port and Harbour Conference. Auckland, New Zealand: Engineers Australia and IPENZ, 2015: 229-234.* (2015).
36. Friedrichs, C. T. Tidal flat morphodynamics: a synthesis. (2011).
37. Alonso, A. C., Van Maren, D. S., Elias, E. P. L., Holthuijsen, S. J. & Wang, Z. B. The contribution of sand and mud to infilling of tidal basins in response to a closure dam. *Mar. Geol.* **439**, 106544 (2021).
38. Van Ledden, M., Van Kesteren, W. G. M. & Winterwerp, J. C. A conceptual framework for the erosion behaviour of sand–mud mixtures. *Cont. Shelf Res.* **24**, 1–11 (2004).

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Author contributions

A.L. conceived the idea of the study, co-worked on the methodology, analysis and visualisation. R.H. developed the methods, and wrote the paper together with A.L. and B.v.M.; A.M. and F.C. co-developed the pixel classification scheme, supported the validation method and conducted the global upscaling. E.K. assisted with incorporating the global shoreline dynamics. S.L. and S.A. supervised initial development of the classification method and reviewed the manuscript.

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